**Table 4.1 Genetic basis of the human ABO blood groups**

| Genotype | ABO blood group phenotype | Antigens present on red blood cell | Antibodies present in blood plasma |
|---|---|---|---|
| $I^AI^A$ | Type A | A | Anti -B |
| $I^AI^o$ | Type A | A | Anti -B |
| $I^BI^B$ | Type B | B | Anti -A |
| $I^BI^o$ | Type B | B | Anti -A |
| $I^AI^B$ | Type AB | A and B | Neither Anti-A nor Anti-B |
| $I^oI^o$ | Type O | Neither A nor B | Anti -A and anti - B |

## Rhesus or Rh Factor

The Rh factor or Rh antigen is found on the surface of erythrocytes. It was discovered in 1940 by Karl Landsteiner and Alexander Wiener in the blood of rhesus monkey, *Macaca rhesus* and later in human beings. The term 'Rh factor' refers to "immunogenic D antigen of the Rh blood group system. An individual having D antigen are Rh D positive ($Rh^+$) and those without D antigen are Rh D negative ($Rh^-$)". Rhesus factor in the blood is inherited as a dominant trait. Naturally occurring Anti D antibodies are absent in the plasma of any normal individual. However if an $Rh^-$ (Rh negative) person is exposed to $Rh^+$ (Rh positive) blood cells (erythrocytes) for the first time, anti D antibodies are formed in the blood of that individual. On the other hand, when an Rh positive person receives Rh negative blood no effect is seen.

# 4.3 Genetic control of Rh factor

## Fisher and Race hypothesis:

Rh factor involves three different pairs of alleles located on three different closely linked loci on the chromosome pair. This system is more commonly in use today, and uses the 'Cde' nomenclature.

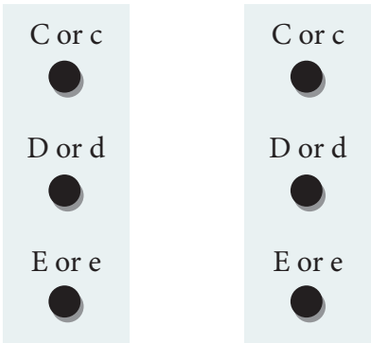



**Fig. 4.1 Fischer and Race hypothesis**

In the above **Fig. 4.1**, three pairs of Rh alleles (Cc, Dd and Ee) occur at 3 different loci on homologous chromosome pair. The possible genotypes will be one C or c, one D or d, one E or e from each chromosome. For e.g. CDE/cde; CdE/cDe; cde/cde; CDe/CdE etc., All genotypes carrying a dominant 'D' allele will produce Rh positive phenotype and double recessive genotype 'dd' will give rise to Rh negative phenotype.

## Wiener Hypothesis

Wiener proposed the existence of eight alleles ($R^1$, $R^2$, $R^0$, $R^z$, r, $r^1$, $r^{11}$, $r^y$) at a single Rh locus. All genotypes carrying a dominant 'R allele' ($R^1$, $R^2$ ,$R^0$ ,$R^z$) will produce Rh positive phenotype and double recessive genotypes (rr, $r^1r^1$, $r^{11}r^{11}$, $r^yr^y$) will give rise to Rh negative phenotype.

## 4.3.1 Incompatibility of Rh Factor – Erythroblastosis foetalis

Rh incompatability has great significance in child birth. If a woman is Rh negative and the man is Rh positive, the foetus may be Rh positive having inherited the factor from



TN_GOVT_XII_Bio_Zoology_EM_chapter 4.indd 47 20-03-2020 10.24.25 AM